Note: Students are required to write the programs along with atleast two outputs for each experiment in the laboratory observation record. The observation record must be signed by the faculty in charge. For every laboratory session, the observation component carries **marks**.

Q1. Write a program that accepts an integer and prints:

- "Positive" if the number is greater than 0
- "Negative" if the number is less than 0
- "Zero" if the number is 0

Solution:

```
n = int(input("Enter an integer: "))
if n > 0:
    print("Positive")
elif n < 0:
    print("Negative")
else:
    print("Zero")
```

Q2..Write a program that reads two integers and prints the larger number.

Solution : Program to find the larger of two numbers

```
num1 = int(input("Enter the first integer: "))
```

```
num2 = int(input("Enter the second integer: "))
if num1 > num2:
    print(num1, "is the larger number.")
else:
    print(num2, "is the larger number.")
```

Sample input and output

```
Enter the first integer: 15
Enter the second integer: 20

20 is the larger number.
```

Q3.Write a program to determine whether a given year is a leap year.

Solution : Program to check whether a year is a leap year

```
year = int(input("Enter a year: "))
if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):
    print(year, "is a Leap Year.")
else:
    print(year, "is Not a Leap Year.")
```

Sample input and output

```
Enter a year: 2024

output:2024 is a Leap Year.
```

Q4.Write a program that prints the multiplication table of a given number up to 10 using a loop.

Program to print the multiplication table of a number
Solution:

```
num = int(input("Enter a number: "))

for i in range(1, 11):
   print(num, "x", i, "=", num * i)
```

Sample input and output

```
Enter a number: 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

Q5.Write a program that reads **N** integers and counts how many are even and how many are odd.

Solution : Program to count even and odd numbers

```
n = int(input("Enter the number of integers: "))
```

```
even_count = 0
odd_count = 0

for i in range(n):
    num = int(input("Enter an integer: "))

    if num % 2 == 0:
        even_count += 1
    else:
        odd_count += 1

print("Even numbers:", even_count)
print("Odd numbers:", odd_count)
```

Sample input and output

Sample input: Enter the number of integers: 5
Enter an integer: 10
Enter an integer: 7
Enter an integer: 4
Enter an integer: 9
Enter an integer: 2

Output: Even numbers: 3
Odd numbers: 2

Q6.Write a program to accept a string and count the number of vowels (a, e, i, o, u) present in it.

Solution : Program to count the number of vowels in a string

```
s = input("Enter a string: ")

count = 0
```

```
for ch in s:
    if ch == 'a' or ch == 'e' or ch == 'i' or ch == 'o' or ch == 'u':
        count = count + 1

print("Number of vowels:", count)
```

Sample input and output

Enter a string: computer
Number of vowels: 3

Q7.Write a program to reverse a given string without using built-in reverse functions.
Solution : Program to reverse a string without using built-in reverse functions

```
s = input("Enter a string: ")

reverse = ""

for ch in s:
    reverse = ch + reverse

print("Reversed string:", reverse)
```

Sample input and output

Enter a string: Python

Reversed string: nohtyP

Q8.Write a program to check whether a given string is a palindrome.

Solution : Program to check whether a string is a palindrome

```
s = input("Enter a string: ")


reverse = ""


for ch in s:
    reverse = ch + reverse


if s == reverse:
    print("Palindrome")
else:
    print("Not a Palindrome")
```

Sample input and output

Enter a string: madam

The string is a palindrome.

Q9.Write a program to count the number of words in a sentence.

Solution: Program to count the number of words in a sentence

```
sentence = input("Enter a sentence: ")


words = 0
in_word = False


for ch in sentence:
    if ch != ' ' and in_word == False:
        words = words + 1
        in_word = True
    elif ch == ' ':
        in_word = False


print("Number of words:", words)
```

Sample input and output

Sample input :Enter a sentence: Python is easy to learn

Number of words: 5

10. **Grade Calculation**

Read marks and display the grade according to the following:

- 90–100 → A
- 80–89 → B
- 70–79 → C
- 60–69 → D
- Below 60 → F

Solution:

```
marks = int(input("Enter marks: "))


if marks >= 90 and marks <= 100:
    print("Grade: A")
elif marks >= 80:
    print("Grade: B")
elif marks >= 70:
    print("Grade: C")
elif marks >= 60:
    print("Grade: D")
else:
    print("Grade: F")
```

Department of CSE(AIML), PES University
UE26CS151A -Python for Computational Problem Solving
Laboratory Week 3

Prepared By,

SONIKA C T